

Where the energy goes on the harvested Apollo4 board

Informed estimates for compute, checkpointing and energy sensing on the AMAP4PEVB Apollo4 Plus with the MP1584EN regulator and 10 mF reservoir. Durations are measured on the board where marked; power is taken from the Apollo4 Plus datasheet with a margin for board overhead. Shunt measurements (V14) will replace these numbers.

Key findings

CHECKPOINTING IS CHEAP

≈ 32 μJ

per typical checkpoint save (25–47 μJ), about a quarter of one LeNet inference. A restore is ≈ 8.6 μJ. One save uses ≈ 0.19% of the 6.2→5.8 V reserve band.

CUSTOM ENGINE VS TFLM

≈ 21% less

energy per inference for the output-stationary engine (tile 8) than TFLM + CMSIS-NN, assuming both run at the same board power, so energy scales with the measured time.

BIGGEST COST: VCAP POLLING

≈ 93%

of a frame's energy is spent reading VCAP: each read takes ≈ 4 ms, and the current firmware reads before every pixel. LPMODE0 reads or polling every 64 pixels would cut frame energy by about 5–13×.

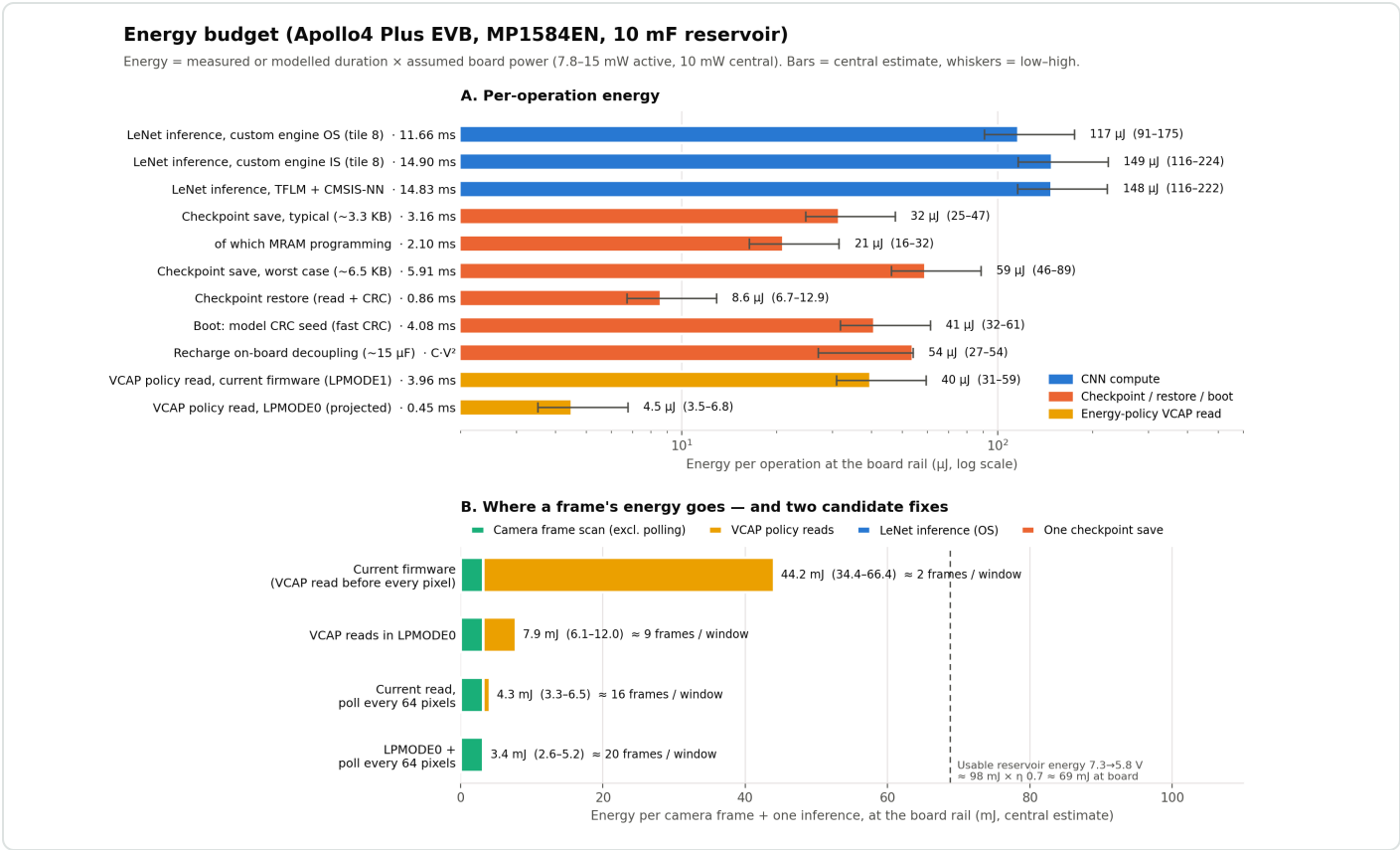


Figure 1. Panel A: per-operation energy at the board rail. Panel B: energy per camera frame plus one inference under the current firmware and three alternatives, against the usable reservoir energy.

How the numbers are built

Every energy figure is a duration multiplied by a power. The durations are mostly measured; the power is the main uncertainty, so each figure carries a low-high range.

$$E \text{ } [\mu\text{J}] = t \text{ } [\text{ms}] \times P \text{ } [\text{mW}] \qquad (1 \text{ ms} \times 1 \text{ mW} = 1 \text{ } \mu\text{J})$$
$$E_{\text{reservoir}} = E_{\text{board}} / \eta \qquad (\eta = \text{MP1584EN efficiency})$$
$$E_{\text{window}} = \frac{1}{2} \cdot C \cdot (V_1^2 - V_2^2) \qquad (C = 10 \text{ mF reservoir})$$

Assumptions

INPUT	LOW	CENTRAL	HIGH	BASIS	TYPE
Board active power, P _{act}	7.8 mW	10 mW	15 mW	Datasheet Table 29: 21.3 μA/MHz × 192 MHz = 4.09 mA × 1.9 V = 7.77 mW (MCU only). Higher values allow for EVB overhead and MRAM programming current.	datasheet + margin
Sleep power during pixel settle	0.34 mW	0.45 mW	0.70 mW	Datasheet ISS2 sleep 180 μA × 1.9 V = 0.34 mW, plus board margin.	datasheet + margin
MP1584EN efficiency at a few mA, η	0.80	0.70	0.60	Typical light-load buck range; not yet measured on this module.	assumption
Operation durations	See the operations table			Board timings at 192 MHz (B. Labonite, Data_tables 09/28) and the 10/01 scope state-DAC capture.	measured
Reservoir and thresholds	10 mF; 7.3 / 6.8 / 6.2 / 5.8 V			Installed hardware and configured firmware thresholds.	configured

Per-operation energy at the board rail (μJ)

OPERATION	DURATION	DURATION SOURCE	LOW	CENTRAL	HIGH	CENTRAL MATH
LeNet inference, custom engine OS (tile 8)	11.66 ms	Board, Data_tables T4	91	117	175	11.66 ms × 10 mW
LeNet inference, custom engine IS (tile 8)	14.90 ms	Board, Data_tables T4	116	149	224	14.90 ms × 10 mW
LeNet inference, TFLM + CMSIS-NN	14.83 ms	Board, Data_tables T4	116	148	222	14.83 ms × 10 mW
Checkpoint save, typical (~3.3 KB)	3.16 ms	Board, Data_tables T16 (mean)	25	32	47	3.16 ms × 10 mW
of which MRAM programming	2.10 ms	Board, Data_tables T16	16	21	32	2.10 ms × 10 mW ≈ 6.3 nJ/byte
Checkpoint save, worst case (~6.5 KB)	5.91 ms	Board, Data_tables T16 (max)	46	59	89	5.91 ms × 10 mW
Checkpoint restore (read + CRC)	0.86 ms	Board, Data_tables T16 (mean)	6.7	8.6	12.9	0.86 ms × 10 mW
Boot: model CRC seed (fast CRC)	4.08 ms	Board, Data_tables T15	32	41	61	4.08 ms × 10 mW
Recharge on-board decoupling	~15 μF	EVB schematic (rev 2)	27	54	54	$\frac{1}{2}CV^2 \dots CV^2$ at 1.9 V
VCAP policy read, current firmware	3.96 ms	Board, scope state DAC (10/01)	31	40	59	3.96 ms × 10 mW
VCAP policy read, LPMODE0 (projected)	0.45 ms	Firmware cost model	3.5	4.5	6.8	0.45 ms × 10 mW

MRAM programming: 2.10 ms for a 3,318-byte record is 0.63 μs per byte, which matches the firmware's 1.5 kB/ms assumption, or about 6.3 nJ per byte at 10 mW. The remaining ≈ 1.1 ms of a save (≈ 11 μJ) is fixed overhead for CRC, packing and the header write. These checkpoint timings come from the earlier engine; the current engine's records are similar in size, but its save path has not been timed yet.

Per-frame energy (mJ at the board rail)

Camera scan = 1,024 px × (3.5 ms × P_sleep + 0.15 ms × P_act)

= 1,024 × (1.58 μJ + 1.50 μJ) ≈ 3.15 mJ

VCAP polling = N_reads × t_read × P_act

= 1,033 × 3.96 ms × 10 mW ≈ 40.9 mJ

Inference = 9.9 ms × 10 mW ≈ 0.10 mJ (OS, 1000-unit band)

One save = 3.16 ms × 10 mW ≈ 0.03 mJ

The 3.5 ms settle per pixel is set in the camera driver (2.5 ms + 1 ms waits, spent in normal sleep). The 0.15 ms of active time per pixel for the AVG16 conversion and mode switching is an estimate.

The firmware calls the VCAP read at the top of every scheduler step, and a camera step is one pixel; that is 1,024 reads per frame, plus 9 per inference in the 1000-unit band. A replay capture that counts the ADC state on the state DAC will confirm this.

FIRMWARE OPTION	SCAN	VCAP POLLING	READS	CNN + SAVE	TOTAL	RANGE	FRAMES / WINDOW
Current firmware: VCAP read before every pixel and CNN quantum	3.15	40.91	1,024 + 9 reads × 3.96 ms	0.13	44.2	34.4–66.4	≈ 2
VCAP reads in LPMODE0	3.15	4.65	1,024 + 9 reads × 0.45 ms	0.13	7.9	6.1–12.0	≈ 9
Current read, poll every 64 pixels	3.15	0.99	16 + 9 reads × 3.96 ms	0.13	4.3	3.3–6.5	≈ 16
LPMODE0 and poll every 64 pixels	3.15	0.11	16 + 9 reads × 0.45 ms	0.13	3.4	2.6–5.2	≈ 20

Reservoir budget

7.3 → 6.8 V : $\frac{1}{2} \cdot 0.01 \cdot (7.3^2 - 6.8^2) = 35.3 \text{ mJ}$

6.8 → 6.2 V : $\frac{1}{2} \cdot 0.01 \cdot (6.8^2 - 6.2^2) = 39.0 \text{ mJ}$

6.2 → 5.8 V : $\frac{1}{2} \cdot 0.01 \cdot (6.2^2 - 5.8^2) = 24.0 \text{ mJ}$

7.3 → 5.8 V : total = 98.2 mJ

At the board (η 0.7) ≈ 68.8 mJ

A typical save draws ≈ 45 μJ from the reservoir (board energy ÷ 0.7). That is ≈ 0.19% of the 24 mJ band between the 6.2 V save threshold and 5.8 V, and it lowers VCAP by about $E / (C \cdot V) \approx 0.8 \text{ mV}$ at 6 V.

Frames per window ignore harvested input and wait-state power, so they compare the options rather than predict run length.

Limits of these estimates

- Board power is the largest uncertainty. The EVB's other loads (level shifters, LEDs, the power-supply tracker) and any extra MRAM programming current are covered only by the 15 mW upper bound.
- Wait and sleep power, the MP1584EN's quiescent current and the 400 k Ω divider's draw are not included.
- The CNN comparison assumes equal power for both engines. SIMD kernels may draw slightly more current, which would narrow the 21% gap.
- With the oscilloscope attached, VCAP readings run about 5% low (an earth loop through the scope and signal generator). Keep this in mind for any run where the scope is connected.
- Next step: the V14 loop measurement (repeated saves with a skip-write control, read on a DMM and the 1 Ω shunt) replaces the checkpoint numbers with measured ones. The same method gives the VCAP read energy.